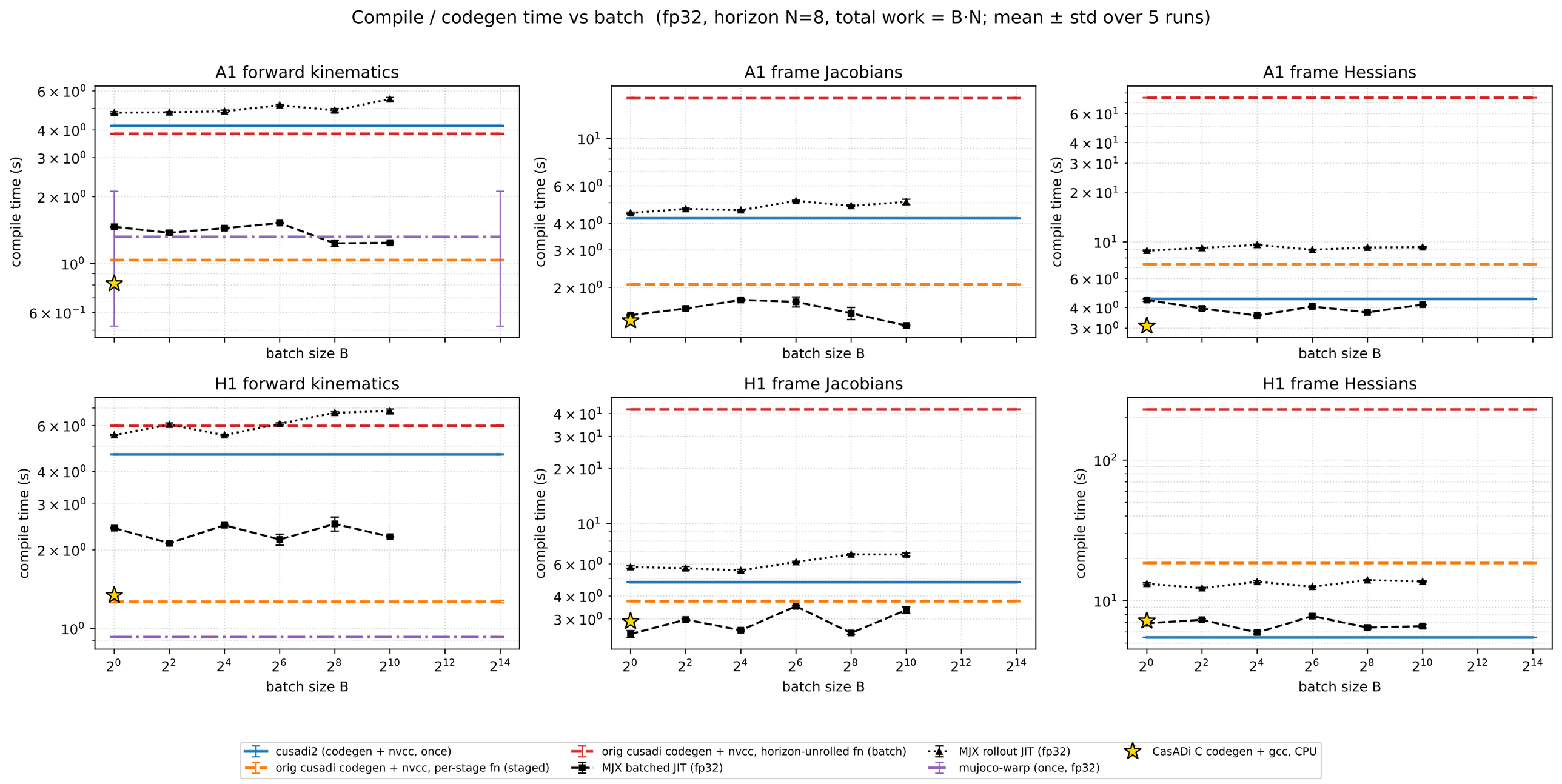
Compile / codegen time vs batch (fp32, horizon N=8, total work = B·N; mean ± std over 5 runs)
A1 forward kinematics
A1 frame Jacobians
A1 frame Hessians
H1 forward kinematics
H1 frame Jacobians
H1 frame Hessians
compile time (s)
batch size B
cusadi2 (codegen + nvcc, once)
orig cusadi codegen + nvcc, per-stage fn (staged)
orig cusadi codegen + nvcc, horizon-unrolled fn (batch)
MJX batched JIT (fp32)
MJX rollout JIT (fp32)
mujoco-warp (once, fp32)
CasADi C codegen + gcc, CPU